

Proposal: Character Relationship Extraction from a Fandom Wiki into a Knowledge Graph

This project will build a simple NLP pipeline to extract character relationships from a Fandom wiki and visualize them as an interactive Knowledge Graph. Character pages will be downloaded through the wiki's Special:Export function as XML files, then parsed, cleaned, and split into sentence-level contexts. Each example will include a page character, another mentioned character, and the surrounding text. The task is to classify their relationship into categories that fit the specific fandom such as family.

The project will compare three relation extraction approaches: a TF-IDF and Logistic Regression baseline, a BiLSTM classifier using token embeddings and entity markers, and a transformer-based classifier such as DistilBERT. Additional experiments may include improved TF-IDF features, such as entity markers and section information.

Models will be evaluated using accuracy, macro-F1, per-class F1, and a small graph-level evaluation. Predicted relationships will then be converted into triples, where characters are nodes and relationships are labeled edges. Finally, interactive PyVis graphs will be generated for each model to compare how different approaches affect the structure and quality of the extracted character network.

Group members

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